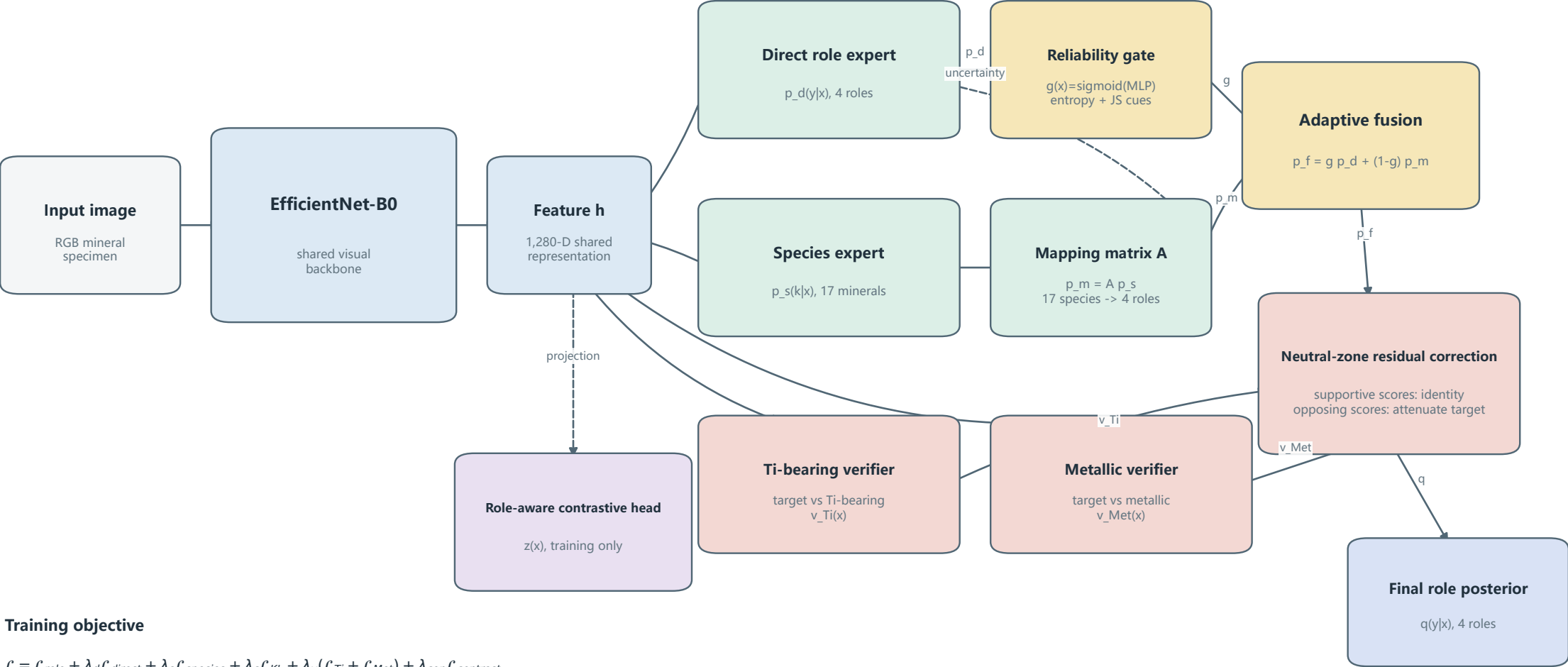


HRGV-Net: hierarchical evidence fusion with hard-negative risk verification

共享视觉表征同时支持角色识别、矿物种类识别、可靠性融合与两类高风险验证



Training objective

$$\mathcal{L} = \mathcal{L}_{role} + \lambda_d \mathcal{L}_{direct} + \lambda_s \mathcal{L}_{species} + \lambda_c \mathcal{L}_{KL} + \lambda_v (\mathcal{L}_{Ti} + \mathcal{L}_{Met}) + \lambda_{con} \mathcal{L}_{contrast}$$

Solid arrows: inference path Dashed arrows: uncertainty/training cues Gradient isolation is evaluated only as an ablation.

$$c_h(v_h) = [\tau_h - v_h]_+ / \tau_h, \quad q_T \propto p_{f,T} \exp[-\beta_{Ti} C_{Ti} - \beta_{Met} C_{Met}], \quad q_j \propto p_{f,j} \ (j \neq T)$$